• MIDTERM PROJECT | 2026 | CREDIT RISK MODELING

HOME CREDIT Default Risk Scoring.

A reproducible end-to-end ML pipeline that turns 7 raw tables into a deployable Probability-of-Default model — with scorecard, risk bands & a Streamlit front-end.

AUTHOR	DOMAIN	COURSE	STACK
Phat	Applied Math + Finance	Credit Risk Modeling	Python · LightGBM · Docker

SECTION 01

THE PROBLEM.

Home Credit serves underbanked borrowers with thin credit files. The business problem: **score each applicant for default risk** using alternative data — bureau, prior applications, installments, POS, credit cards.

01

Why it matters

Default rate is low (~8%) — imbalanced classification; naive accuracy is misleading.

02

Cost asymmetry

Missing a defaulter is far more expensive than rejecting a good payer.

03

Calibrated PD

Calibrated PD feeds downstr
expected loss, and loan pricir

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01 · PROBLEM & MOTIVATION

Deliverables

Six concrete artifacts shipped with this project — every chart and number in this deck is reproducible from the source code.

MODEL ARTIFACTS

- Trained & calibrated model `model.joblib`
- Metrics table `model/metrics_table.csv`
- Per-customer PD + rating `predictions_test.csv`
- Model card `model_card.json`

DEPLOYMENT & DOCS

- Streamlit web app (4 tabs) + Dockerfile
- End-to-end notebook `home_credit_pipeline.ipynb`
- This slide deck `slides.pdf`
- Power BI dashboard `HomeCreditRiskDashboard.pbix`

TRAIN ROWS	TEST ROWS	FEATURES	DEFAULT RAT
5,250	7,500	301	8.0%

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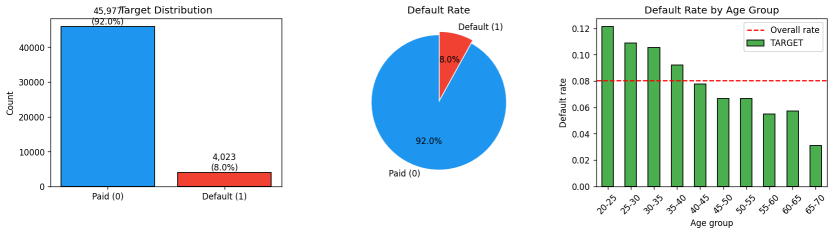
02 · DATASET

Home Credit Default Risk — Kaggle

`application_train.csv` is the master table (one row per customer) with the TARGET label. Six auxiliary tables are aggregated to `SK_ID_CURR` to enrich the fe
set.

TARGET DISTRIBUTION



7 SOURCE TABLES

TABLE	ROWS	U
application_train	**307,511**	M
bureau	1.7M	O
bureau_balance	27.3M	M
previous_application	1.7M	Pr
installments_payments	13.6M	Re
POS_CASH_balance	10.0M	P
credit_card_balance	3.8M	C

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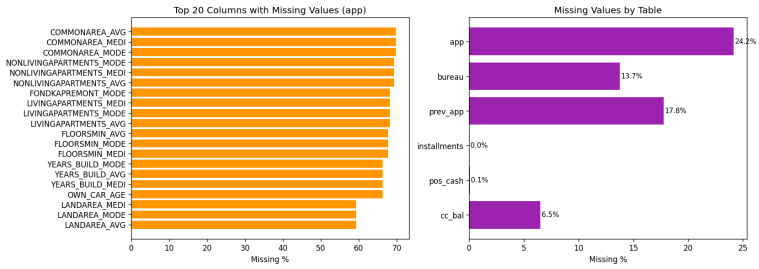
03 · EXPLORATORY DATA ANALYSIS

EDA · Target & Missing Values

Class imbalance and data quality drive downstream modeling choices. Top 20 columns by missing-rate are the most informative signals.

TOP 20 COLUMNS BY MISSING RATE



KEY FINDINGS

- **Default rate ≈ 8%** — heavily imbalanced, naive accuracy ≈ 91.9
- **20–25 cohort** is the riskiest age group; 35–55 cohort is safest.
- **~50+ columns** have meaningful missingness, dominated by ho
- These shape the **imputation & feature-engineering** strategies
- **EXT_SOURCE_*** columns are sparse but most predictive — ca

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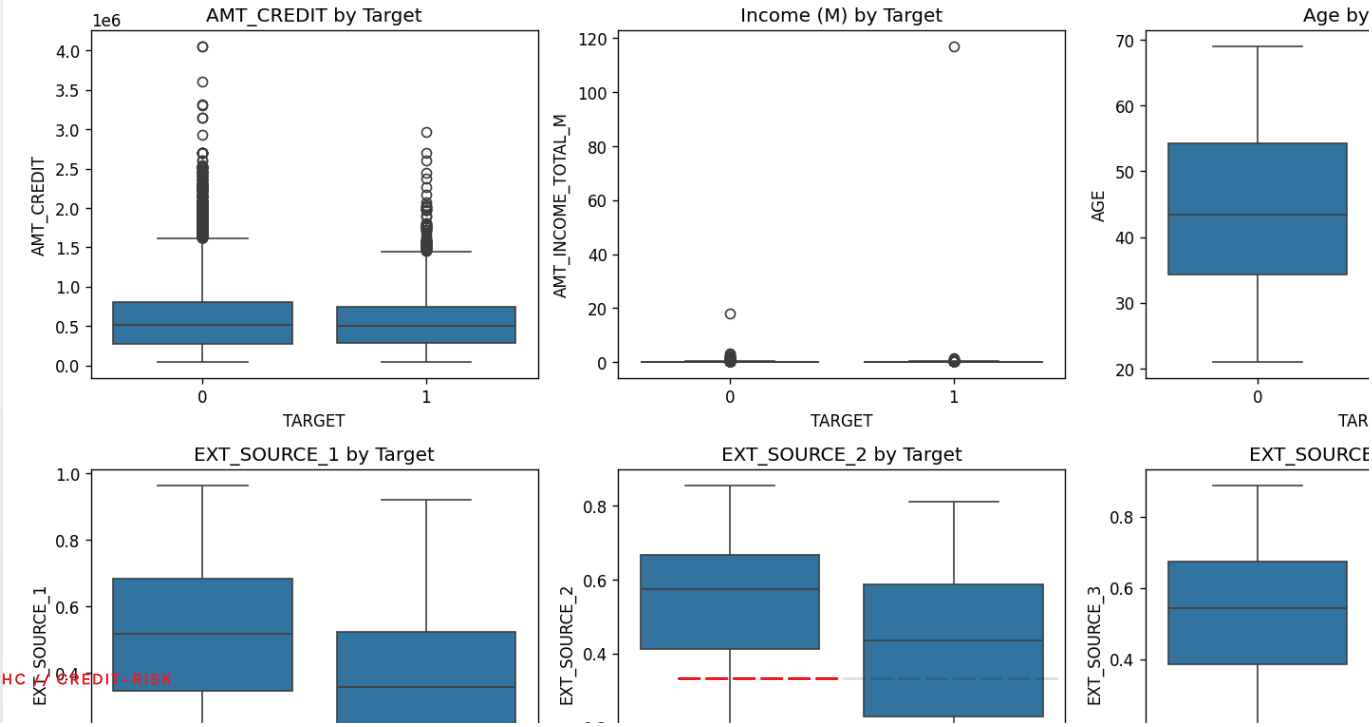
04 · EDA

EDA · Key Features by Target

Comparing distributions of `AMT_CREDIT`, `AMT_INCOME`, `AGE` and the three `EXT_SOURCE` external scores between defaulters and non-defaulters.



05 · FEATURE ENGINEERING

Feature Engineering Pipeline

Starting from 122 raw columns, we engineer **301 features** across 4 functional blocks.

01 · APPLICATION CLEANING

- `DAYS_EMPLOYED` sentinel (365243) → NaN + flag
- `CODE_GENDER` XNA → NaN
- `OWN_CAR_AGE` = NaN if no car
- Time: `AGE_YEARS, EMPLOYED_YEARS, REGISTERED_YEARS`
- Ratios: `CREDIT_INCOME, ANNUITY_INCOME, CREDIT_TERM, CREDIT_GOODS, INCOME_PER_FAM`
- `EXT_SOURCE` aggregates: MEAN / MIN / MAX / STD / PROD / COUNT
- `NUM_DOCS_SUBMITTED`

02 · BUREAU AGGREGATION

- Group by `SK_ID_CURR`
- `BUR_ACTIVE_CNT, BUR_OVERDUE_CNT`
- `BUR_CREDIT_SUM, BUR_DEBT_SUM, BUR_OVERDUE_AMT`
- `BUR_DEBT_RATIO = debt / credit`
- `BUR_LOAN_CNT, BUR_DAYS_CREDIT_MIN`
- `BUR_TYPE_*` one-hot (per credit type)
- `HAS_BUREAU` flag

03 · PREVIOUS APPS + INSTALLMENTS

- `PREV_AMT_SUM/MEAN, PREV_CREDIT_MAX`
- `PREV_DAYS_DEC_MIN, PREV_CNT_PAYMENT_MEAN`
- `PREV_APPROVE_RATE`
- `INST_PAY_RATIO, INST_LATE_MEAN`
- `INST_LATE_7D_SUM`

04 · POS_CASH + CREDIT CARD

- `POS_UTIL_MEAN, POS_CNT`
- `CC_UTIL_MEAN`
- Per-customer aggregations, merged to master
- Final **301-dim feature vector** per customer

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06 · METHODOLOGY

Methodology · Modeling Pipeline

We compare **4 models**: 2 logistic-regression baselines (unweighted / class-balanced) and 2 gradient-boosting trees (raw + isotonic-calibrated). Optuna (15 tunes LightGBM hyperparameters; early stopping on the validation fold.

STEP	OPERATION	DETAIL
1	Split	Train / Valid / Test (70 / 15 / 15) stratified on TARGET, fixed seed
2	Preprocess	ColumnTransformer: median impute (num) + most-freq (cat) + one-hot (cat, drop-first) + standardize (num)
3	Baseline	Logistic Regression (LR) + LR-balanced (`class_weight='balanced'`)
4	Tuned model	LightGBM (Optuna 15 trials) — `n_estimators=205, num_leaves=53, lr=0.047`
5	Calibration	Isotonic regression on validation — Brier pre=0.0677 → post=0.0678
6	Threshold	F1-driven (recall ≥ 80%) — `τ* = 0.05`
7	Evaluate	AUC, Gini, KS, Brier, PR-AUC, F1 — test set, 7,500 unseen customers

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07 · RESULTS

Results · Test-Set Comparison (n = 7,500)

All four models evaluated on the same held-out test fold. **LightGBM raw** has the best AUC; the calibrated variant is selected for deployment because its prob
are trustworthy for downstream scorecarding.

MODEL	AUC	GINI	KS	BRIER	PR-AUC	F1	RECALL
Logistic	0.7452	0.4904	0.3689	0.0684	0.2350	0.2248	0.7927
Logistic_balanced	0.7412	0.4823	0.3626	**0.1940**	0.2288	0.1535	1.0000
LightGBM	**0.7594**	**0.5188**	**0.3858**	**0.0677**	**0.2365**	**0.2294**	0.8375
LightGBM_calibrated ✓	0.7582	0.5164	0.3751	0.0678	0.2282	0.2232	**0.8624**

AUC
0.758

GINI
0.516

KS
0.375

RECALL @ T=
86.2%

SELECTED MODEL

LightGBM + Isotonic Calibration. Catches **86.2% of defaulters** at the cost of flagging ~12.8% precision.

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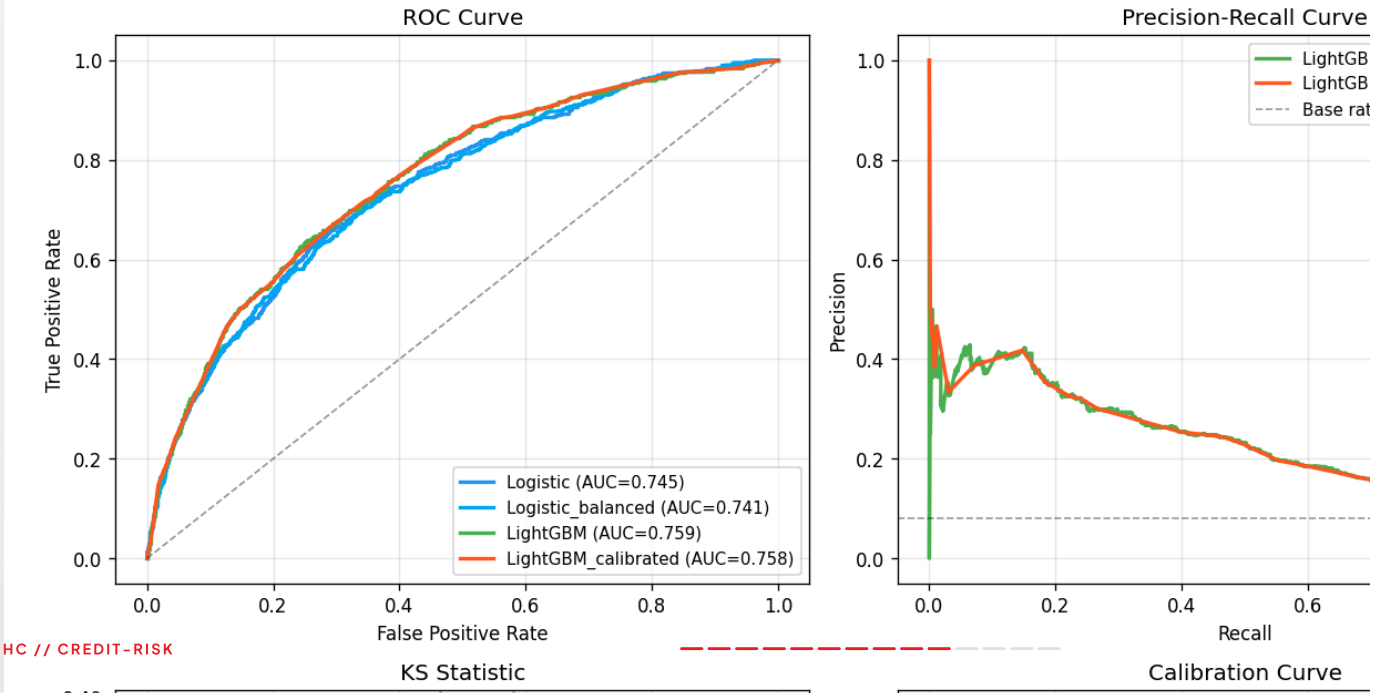
08 · EVALUATION PLOTS

Evaluation Plots · ROC · PR · KS · Calibration

Composite chart: ROC (discrimination) · PR (under class imbalance) · KS (separation) · Calibration (reliability). The isotonic step brings predicted PD close to e
rate.



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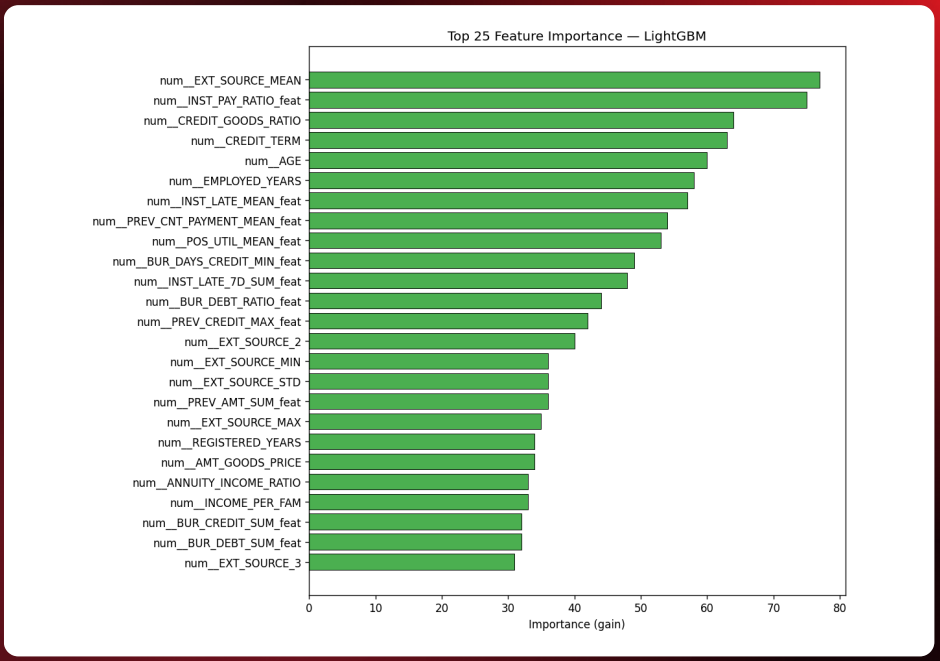
09 · FEATURE IMPORTANCE

Top-25 Feature Importance · LightGBM (gain)

Top features come from engineered ratios, installment history, and external scores — the model relies on behavioural signals, not raw demographics.



#	FEATURE
1	PREV_CNT_PAYMENT_MEAN
2	CREDIT_TERM
3	EXT_SOURCE_MEAN
4	CREDIT_GOODS_RATIO
5	INST_PAY_RATIO
6	POS_UTIL_MEAN
7	EMPLOYED_YEARS
8	BUR_DEBT_RATIO
9	PREV_APPROVE_RATE
10	ANNUITY_INCOME_RATIO

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10 · SCORECARD & RISK BANDS

From PD to Score · Scorecard & Risk Bands

We translate PD into a credit score (Siddiqui 2013 convention: base = 600, PDO = 20, odds = 50:1) and bucket into 5 rating bands.

▸ SCORE FORMULA

$$score(PD) = 712.88 - 28.8539 \cdot \ln\left(\frac{PD}{1 - PD}\right)$$

RATING	SCORE RANGE	AVG PD (TEST)	EMPIRICAL DEFAULT RATE
A (Low risk)	760 – 1000	1.79%	1.92%
B (Acceptable)	640 – 760	4.67%	5.02%
C (Watch-list)	480 – 640	8.47%	8.70%
D (Sub-standard)	360 – 480	14.89%	14.84%
E (High risk)	0 – 360	**31.59%**	**30.04%**

▸ VALIDATION

Empirical default rate rises monotonically across bands — the scorecard is well-calibrated.

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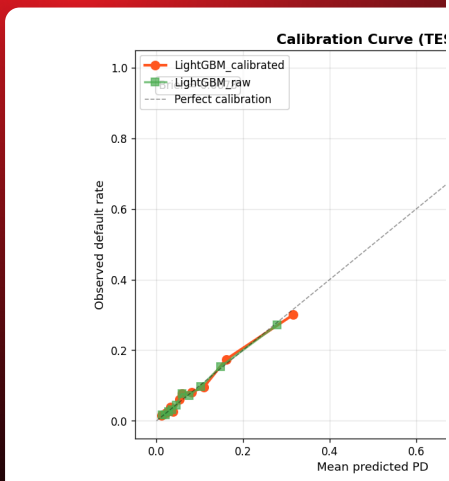
11 · BUSINESS IMPACT

Business Impact · What does 86.2% recall buy?

At `τ = 0.05`, on 7,500 test customers (603 defaulters), the model flags 603 × 0.862 ≈ **520 defaulters** for manual review. Compare to the 8% base rate:

RATING	POP.	DEFAULTERS	CAPTURE	LIFT
Band A	2,185	42	7.0%	0.24×
Band B	2,372	119	19.7%	0.62×
Band C	1,241	108	17.9%	1.08×
Band D	1,166	173	28.7%	1.85×
Band E	**536**	**161**	**26.7%**	**3.74×**

CALIBRATION CURVE



RECALL @ T=0.05
86.2%

PRECISION
12.8%

DEFAULTERS CAUGHT
520 / 603

AUC
0.758

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12 · LIMITATIONS & FUTURE WORK

Limitations & Future Work

Honest assessment of what this pipeline does and does not yet cover.

LIMITATIONS

- **Sample size** — notebook executed on a 12k-row subsample to fit local compute. Re-running on the full 307k rows is expected to lift AUC further.
- **Macro recall vs precision** — at τ=0.05 we prioritise catching defaulters; a higher τ would improve precision at the cost of recall. Policy choice, not a model flaw.
- **Drift monitoring** — Home Credit portfolio evolves yearly; PSI / KS drift on EXT_SOURCE_*, AMT_* should be tracked quarterly.
- **Fairness audit** — disaggregated metrics (CODE_GENDER, AGE_BAND) required before production rollout.

FUTURE WORK

- **Rejected-inference** — current training data only includes approv inference population would close the selection-bias gap.
- **Deep tabular models** — TabNet, FT-Transformer, and other deep architectures are next steps.
- **Time-series of installment history** — sequence models (LSTM, Ti deltas can extract additional signal.
- **Calibration refresh** — quarterly re-fit of isotonic regression as pc

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THANK YOU.

Questions, feedback, collaboration — always welcome.

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github.com/phat/hc-credit-risk

streamlit.hc-credit-risk.app

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